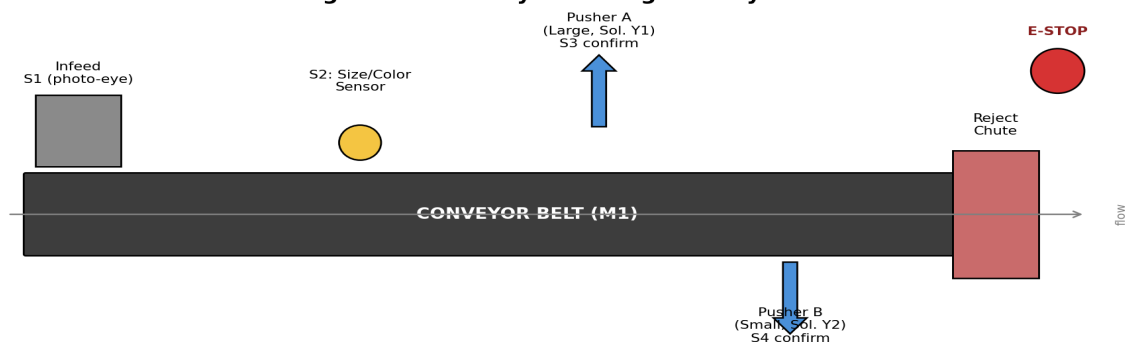


PROJECT 4

PLC-Based Conveyor Sorting System

Design of Industrial Control Logic for an Automated
Box Sorting Line — Size & Color Classification

Figure 2 — Conveyor Sorting Cell Layout



Discipline: Robotics & Automation — Industrial Controls

Deliverable: Ladder Logic / State-Machine Control Design + Documentation

Key Skills Demonstrated: PLC Programming, Ladder Logic, Sensor Integration, Industrial Safety Standards

1. Project Overview & Goal

This project defines the complete industrial control logic for a PLC-driven conveyor sorting cell. Boxes travel along a single infeed conveyor and are classified by **size** and **color** using a photoelectric/vision sensor pair, then diverted to one of three destinations — a large-item pusher lane, a small-item pusher lane, or a reject chute for unclassified items — using pneumatic pushers. The design below covers I/O mapping, the sequencing logic (ladder logic and an equivalent state machine), safety interlocking, and a validation/testing plan suitable for commissioning on a real PLC (e.g., Allen-Bradley CompactLogix, Siemens S7-1200, or a soft-PLC such as OpenPLC).

- **Primary objective:** sort boxes by size or color on an automated conveyor line using safe, deterministic PLC logic.
- **Design deliverables:** I/O map, ladder logic diagram, structured-text equivalent, state machine, E-Stop / safety interlock scheme, and a test procedure.
- **Assumed hardware:** 1 conveyor motor (VFD-driven), 2 pneumatic pusher solenoids, 4 sensors, 1 E-Stop, Start/Stop/Reset pushbuttons, and a stack light.

2. System Description

The mechanical cell consists of a single belt conveyor (M1) that carries boxes past an infeed presence sensor (S1), a combined size/color classification sensor (S2), and two pneumatic pusher stations (Pusher A for large items, Pusher B for small items). Items that fail classification continue to a gravity reject chute at the end of the line. Confirmation sensors (S3, S4) verify that each pusher has completed its stroke before the sequence resumes.

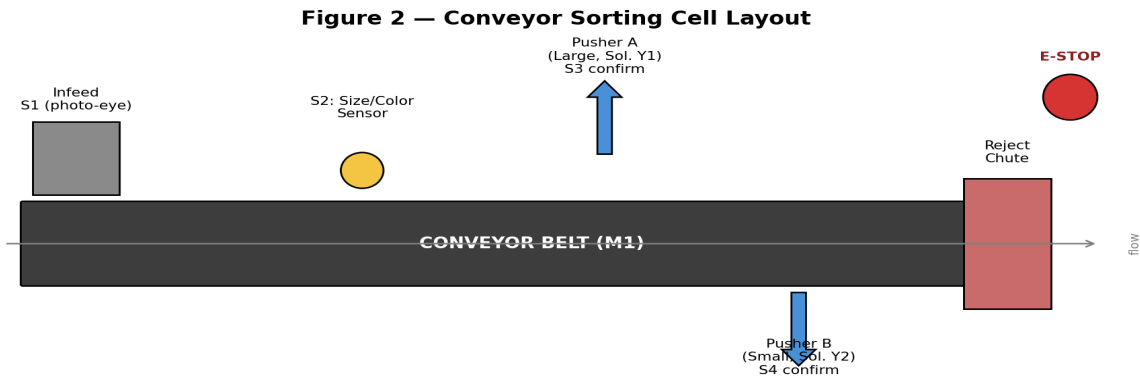


Figure 2 — Physical layout of the conveyor sorting cell.

3. I/O Mapping

3.1 Digital Inputs

Address	Symbol	Device	Description
I:0/0	PB_START	Pushbutton (NO)	Starts the conveyor / system
I:0/1	PB_STOP	Pushbutton (NC)	Normal stop request

I:0/2	PB_RESET	Pushbutton (NO)	Clears a latched fault after E-Stop is released
I:0/3	ESTOP_OK	E-Stop relay (NC contact)	TRUE = E-Stop NOT pressed (healthy)
I:0/4	S1_INFEED	Photoelectric sensor	Box present at infeed
I:0/5	S2_PRESENT	Photoelectric sensor	Box present at classification station
I:0/6	S2_SIZE_LG	Size sensor output	TRUE = box classified as Large
I:0/7	S2_COLOR_OK	Color sensor output	TRUE = box color matches accepted spec
I:1/0	S3_PUSHER_A_HOME	Proximity sensor	Pusher A retracted (home) position
I:1/1	S3_PUSHER_A_EXT	Proximity sensor	Pusher A extended position
I:1/2	S4_PUSHER_B_HOME	Proximity sensor	Pusher B retracted (home) position
I:1/3	S4_PUSHER_B_EXT	Proximity sensor	Pusher B extended position
I:1/4	AIR_PRESS_OK	Pressure switch	Pneumatic supply pressure adequate
I:1/5	M1_OVERLOAD_OK	Motor overload (NC contact)	TRUE = no motor overload fault

3.2 Digital Outputs

Address	Symbol	Device	Description
O:0/0	M1_RUN	Motor contactor / VFD run	Runs conveyor belt forward
O:0/1	Y1_PUSHER_A	Solenoid valve	Extends Pusher A (large-item diverter)
O:0/2	Y2_PUSHER_B	Solenoid valve	Extends Pusher B (small-item diverter)
O:0/3	LT_RUN_GRN	Stack light (green)	System running normally
O:0/4	LT_FAULT_RED	Stack light (red)	Fault / E-Stop active
O:0/5	LT_IDLE_AMB	Stack light (amber)	System idle / stopped, no fault
O:0/6	HORN	Audible alarm	Pulses briefly before belt restart

4. Control Sequence — State Machine

The sequencing logic is implemented as a step/state machine (equivalent to an SFC — Sequential Function Chart). Each state corresponds to a control word value held in an internal register (e.g., N7:0 or a Siemens DINT tag SortState), which keeps the ladder logic simple and scan-time efficient compared to a long chain of latched coils.

Figure 1 — Conveyor Sorting State Machine

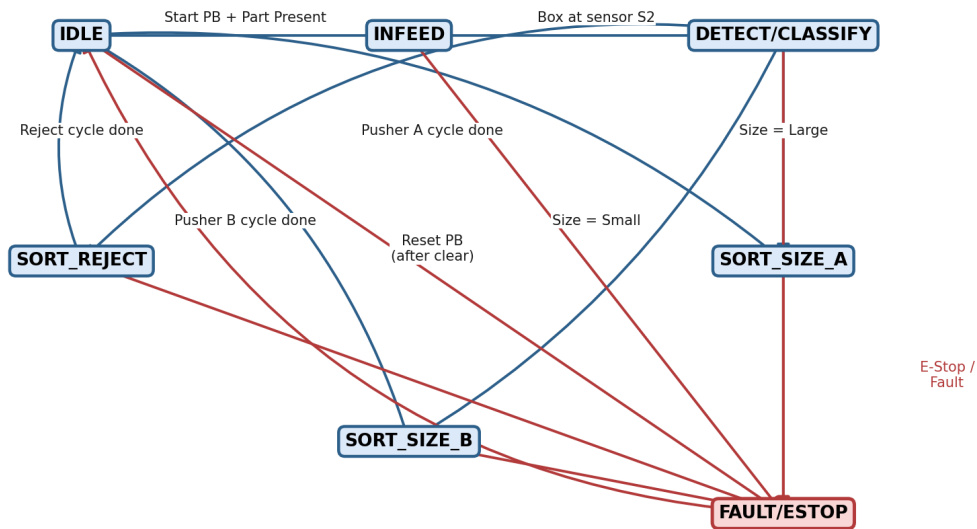


Figure 1 — State transition diagram for the sorting sequence. Any state can transition immediately to FAULT/ESTOP.

State	Description
IDLE	Belt stopped or waiting. Waits for Start PB and a box present at S1.
INFEED	Belt running; box travels from S1 toward the classification sensor S2.
DETECT/CLASSIFY	Belt continues; box size (S2_SIZE_LG) and color (S2_COLOR_OK) are latched when S2_PRESENT rises.
SORT_SIZE_A	Large, color-OK box reaches Pusher A window; Y1 extends, confirms via S3_EXT, then retracts (S3_HOME).
SORT_SIZE_B	Small, color-OK box reaches Pusher B window; Y2 extends, confirms via S4_EXT, then retracts (S4_HOME).
SORT_REJECT	Color mismatch: no pusher fires; box continues to the end-of-line reject chute.
FAULT/ESTOP	E-Stop pressed, air pressure low, or motor overload. All outputs de-energized immediately.

5. Ladder Logic Design

Below is the rung-by-rung ladder logic, written in standard IEC 61131-3 notation (contacts —| |—, coils —()—). The state register `SortState` (INT) drives the sequence; each rung is gated by an equality comparison on that register so only one branch of the machine is active per scan.

5.1 Rung Diagram (text representation)

```
RUNG 0 -- MASTER SAFETY / RUN PERMISSIVE
  ESTOP_OK   AIR_PRESS_OK   M1_OVERLOAD_OK               (SYS_OK)
  --| |-----| |-----| |------( )--

RUNG 1 -- START / STOP LATCH (Seal-in)
  PB_START    SYS_OK    PB_STOP                (SYS_RUN)
  --| |-----| |-----|/|------( )--
      SYS_RUN
      --| |-----+-----+
  NOT SYS_OK -----( RESET SYS_RUN )

RUNG 2 -- STATE 0: IDLE -> INFEED
  SortState=0   SYS_RUN   S1_INFEED
  --[EQU]-----| |-----| |-----[MOV 1 -> SortState]--

RUNG 3 -- BELT RUN OUTPUT
  SYS_RUN      SortState<=>0(Fault excluded)            (M1_RUN)
  --| |------( )--

RUNG 4 -- STATE 1: INFEED -> DETECT/CLASSIFY
  SortState=1    S2_PRESENT
  --[EQU]-----| |-----[MOV 2 -> SortState]--
                          [MOV S2_SIZE_LG -> BoxSizeLg]
                          [MOV S2_COLOR_OK -> BoxColorOk]

RUNG 5 -- STATE 2: CLASSIFY -> branch
  SortState=2    BoxColorOk    BoxSizeLg
  --[EQU]-----| |-----| |-----[MOV 3 -> SortState]--    (-> SORT_SIZE_A)
  SortState=2    BoxColorOk    BoxSizeLg(NC)
  --[EQU]-----| |-----|/|-----[MOV 4 -> SortState]--    (-> SORT_SIZE_B)
  SortState=2    BoxColorOk(NC)
  --[EQU]-----|/|-----[MOV 5 -> SortState]--    (-> SORT_REJECT)

RUNG 6 -- STATE 3: SORT_SIZE_A (Pusher A sequence)
  SortState=3
  --[EQU]------( )--
  S3_PUSHER_A_EXT -----[MOV 0 -> SortState]--
      (retract confirmed by S3_PUSHER_A_HOME before next box admitted)

RUNG 7 -- STATE 4: SORT_SIZE_B (Pusher B sequence)
  SortState=4
  --[EQU]------( )--
  S4_PUSHER_B_EXT -----[MOV 0 -> SortState]--

RUNG 8 -- STATE 5: SORT_REJECT (no pusher; timed pass-through)
  SortState=5 --[TON T1, PT=2.0s]-----DN-----[MOV 0 -> SortState]--

RUNG 9 -- STACK LIGHT / HORN
  SYS_RUN -----(LT_RUN_GRN)--
  NOT SYS_RUN NOT Fault -----(LT_IDLE_AMB)--
  Fault -----(LT_FAULT_RED)--
```

5.2 Equivalent Structured Text (IEC 61131-3)

```
(* Master safety permissive *)
SYS_OK := ESTOP_OK AND AIR_PRESS_OK AND M1_OVERLOAD_OK;

(* Start/Stop seal-in *)
IF (PB_START AND SYS_OK) OR SYS_RUN THEN
    SYS_RUN := SYS_OK AND NOT PB_STOP;
ELSE
    SYS_RUN := FALSE;
END_IF;

(* Belt runs whenever system running and not mid-fault *)
M1_RUN := SYS_RUN;

CASE SortState OF
    0: (* IDLE *)
        IF SYS_RUN AND S1_INFEED THEN SortState := 1; END_IF;

    1: (* INFEED *)
        IF S2_PRESENT THEN
            BoxSizeLg := S2_SIZE_LG;
            BoxColorOk := S2_COLOR_OK;
            SortState := 2;
        END_IF;

    2: (* CLASSIFY *)
        IF BoxColorOk AND BoxSizeLg THEN SortState := 3; (* -> Pusher A *)
        ELSIF BoxColorOk AND NOT BoxSizeLg THEN SortState := 4; (* -> Pusher B *)
        ELSE SortState := 5; (* -> Reject *)
        END_IF;

    3: (* SORT_SIZE_A *)
        Y1_PUSHER_A := TRUE;
        IF S3_PUSHER_A_EXT THEN
            Y1_PUSHER_A := FALSE;
            IF S3_PUSHER_A_HOME THEN SortState := 0; END_IF;
        END_IF;

    4: (* SORT_SIZE_B *)
        Y2_PUSHER_B := TRUE;
        IF S4_PUSHER_B_EXT THEN
            Y2_PUSHER_B := FALSE;
            IF S4_PUSHER_B_HOME THEN SortState := 0; END_IF;
        END_IF;

    5: (* SORT_REJECT -- timed pass-through, no pusher fires *)
        T1(IN := TRUE, PT := T#2s);
        IF T1.Q THEN SortState := 0; T1(IN := FALSE); END_IF;
END_CASE;

(* E-Stop / fault overrides everything, every scan *)
IF NOT SYS_OK THEN
    M1_RUN := FALSE;
    Y1_PUSHER_A := FALSE;
    Y2_PUSHER_B := FALSE;
    SortState := 0; (* forces re-init once PB_RESET is pressed and SYS_OK returns *)
END_IF;
```

6. Safety Interlocks

Safety logic is implemented independently of the sequencing state machine so a fault in the sort logic can never mask an E-Stop condition. The design follows a de-energize-to-stop philosophy: any loss of the safety permissive immediately drops every output, regardless of what state the sequencer is in.

- **Hardwired E-Stop:** the E-Stop pushbutton is wired through a safety relay with forced-guided contacts, hard-wired directly into the motor contactor and solenoid valve power feeds — not solely through PLC logic — so the belt and pushers stop even if the PLC scan is frozen.
- **Software permissive (SYS_OK):** mirrors the E-Stop, air-pressure, and motor-overload status into the PLC for sequencing and diagnostics, per Rung 0 / the ST SYS_OK expression above.
- **Immediate output kill on fault:** the fault branch executes every scan (not just on transition) and forces M1_RUN, Y1_PUSHER_A, and Y2_PUSHER_B false, and resets SortState to 0.
- **Manual, deliberate restart:** the system will not auto-resume after a fault clears. An operator must physically release the E-Stop and press the dedicated Reset pushbutton, then Start again — preventing an unattended, unexpected belt restart.
- **Pre-start horn/light warning:** the horn (O:0/6) pulses for a short interval before M1_RUN energizes after any stop, per common industrial practice (e.g., NFPA 79 general guidance on machine safeguarding) so nearby personnel are alerted before motion resumes.
- **Pneumatic safety:** pushers will not fire (AIR_PRESS_OK gates SYS_OK) if supply pressure is below the setpoint, preventing a weak, unpredictable half-stroke.
- **Confirmed actuator position:** the sequencer will not admit the next box until the active pusher has confirmed both extend and retract via its proximity sensors, preventing a stuck cylinder from silently jamming the line.

Note: This document is an educational design exercise. A real installation must be reviewed against the applicable machine-safety standards for the jurisdiction and application (e.g., ISO 13849-1, IEC 62061, NFPA 79, OSHA 1910 Subpart O) and signed off by a qualified safety engineer before commissioning.

7. Testing & Validation Plan

Commissioning proceeds from the safety circuit outward to the full sequence, so that no sorting logic is exercised until the safety chain is proven.

#	Test	Expected Result
1	Press E-Stop with system idle	SYS_OK and SYS_RUN drop; red stack light on; belt cannot start
2	Press E-Stop while belt running	M1_RUN, Y1, Y2 de-energize within one scan; SortState resets to 0
3	Release E-Stop without pressing Reset	System remains stopped (no auto-restart)
4	Press Reset, then Start	Horn pulses briefly, then M1_RUN energizes; green light on
5	Feed a large, correct-color box	Pusher A fires once box reaches station; box diverted to Lane A
6	Feed a small, correct-color box	Pusher B fires; box diverted to Lane B
7	Feed an incorrect-color box	No pusher fires; box passes to reject chute
8	Drop supply air pressure below setpoint	SYS_OK drops; pushers disabled; belt stops safely
9	Simulate motor overload trip	SYS_OK drops; fault light on; sequence halts safely

10	Jam a pusher (block S3/S4 extend)	Sequence holds state (no next-box admission) until cleared
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8. Conclusion

This design satisfies the project's key requirements: a full digital I/O map covering sensors, motors, and pneumatic pushers; an industrial-style sequencing solution expressed both as ladder logic and as an equivalent structured-text state machine; and a safety interlock scheme in which the E-Stop path is independent of, and takes priority over, the sorting sequence. The state-machine approach (rather than a long chain of latch/unlatch rungs) keeps the logic compact, easy to troubleshoot on the plant floor, and straightforward to extend — for example, adding a third size class or a barcode-based routing scheme would only require new CASE branches rather than a redesign of the safety or I/O layers.

8.1 Key Skills Demonstrated

- **PLC programming fundamentals:** digital I/O addressing, scan-cycle-safe logic, and internal state registers.
- **Ladder logic:** contact/coil rungs, latching seal-ins, timers, and comparison instructions.
- **Sensor integration:** photoelectric presence sensing, classification sensors, and proximity-based actuator confirmation.
- **Industrial safety standards:** de-energize-to-stop design, hardwired E-Stop chains, manual-reset restart discipline, and pre-motion warning devices.

*Prepared as an educational systems-design exercise for Project 4: PLC-Based Conveyor Sorting System
(Robotics & Automation curriculum).*